ANNA DRISHTI

DETECT · PREDICT · PROTECT

SEE DEEPER.
SAVE GRAIN.

Deep-Stack Grain Monitoring
& Early-Warning System

SMART INDIA HACKATHON 2026

Agriculture, Food Technology & Rural Development

THE PROBLEM

SPOILAGE STARTS WHERE INSPECTION STOPS.

1. LOCALIZED MOISTURE

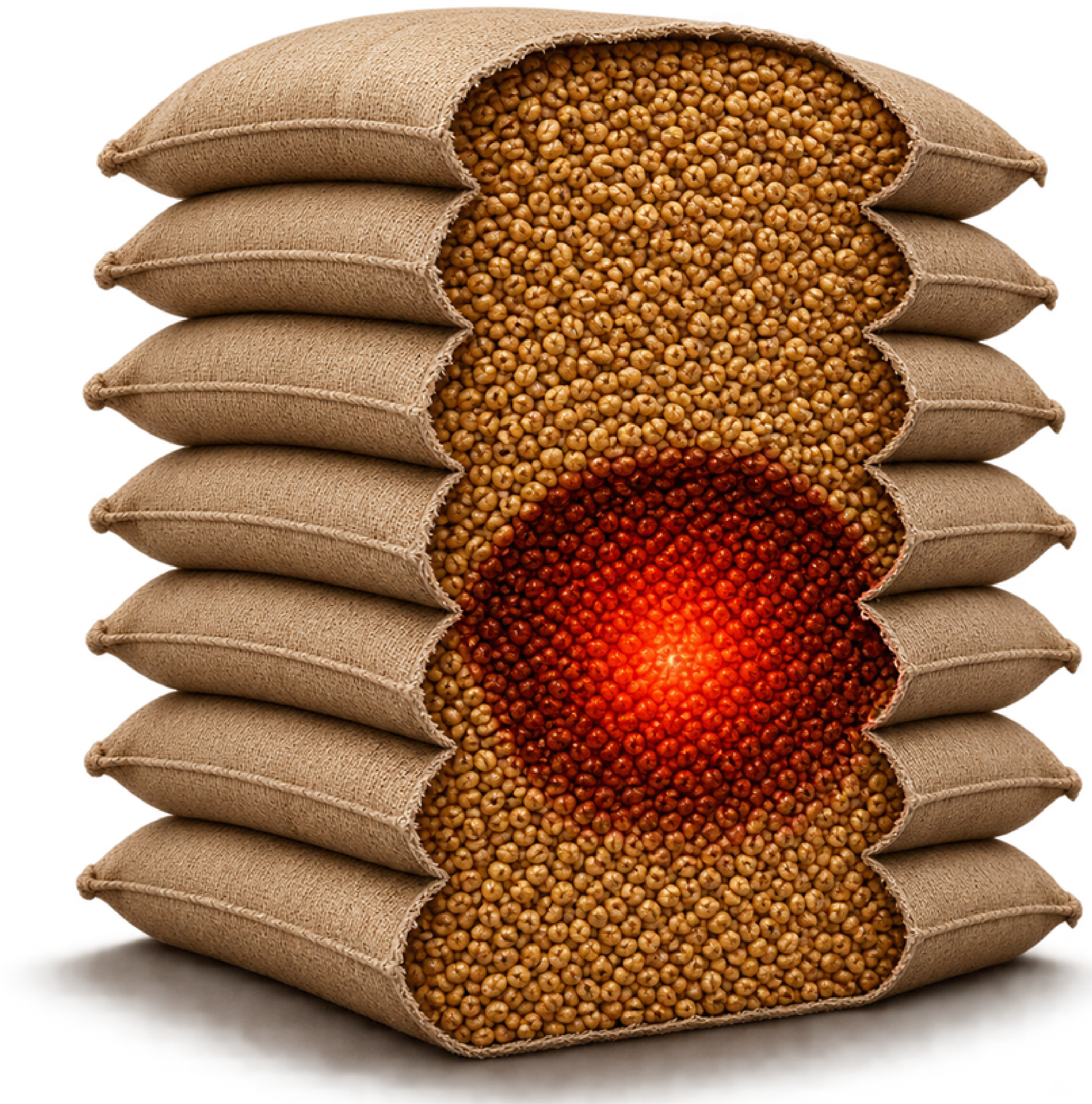
Roof leaks, wet bags, condensation or wet corners create a small **wet pocket** inside a grain stack.

2. HIDDEN DETORIATION

Higher moisture / water activity creates favourable conditions for **fungal growth** and grain detoriation.

3. DAMAGE SPREADS SILENTLY

Dry matter is lost and, in cases involving Aspergillus flavus, **aflatoxin risk** may develop before visible mould appears.



Indian cereal post-harvest losses assessed at 3.89–5.92% — NABCONS study reported to Parliament, 2024

1 Conventional inspection can sample what is reachable - but deterioration can begin where it cannot.

2 **THE REAL CHALLENGE:** Detecting hidden deterioration deep inside grain stacks.

We need a way to SEE DEEPER. BEFORE IT SPREADS.

THE INSIGHT

SPOILAGE BREATHES
GAS ARRIVES FIRST.

1

SPOILAGE RESPIRES

Grain is alive, and so are the fungi on it. Both consume sugar and oxygen and **release CO_2, water and heat** — from a single reaction.

2

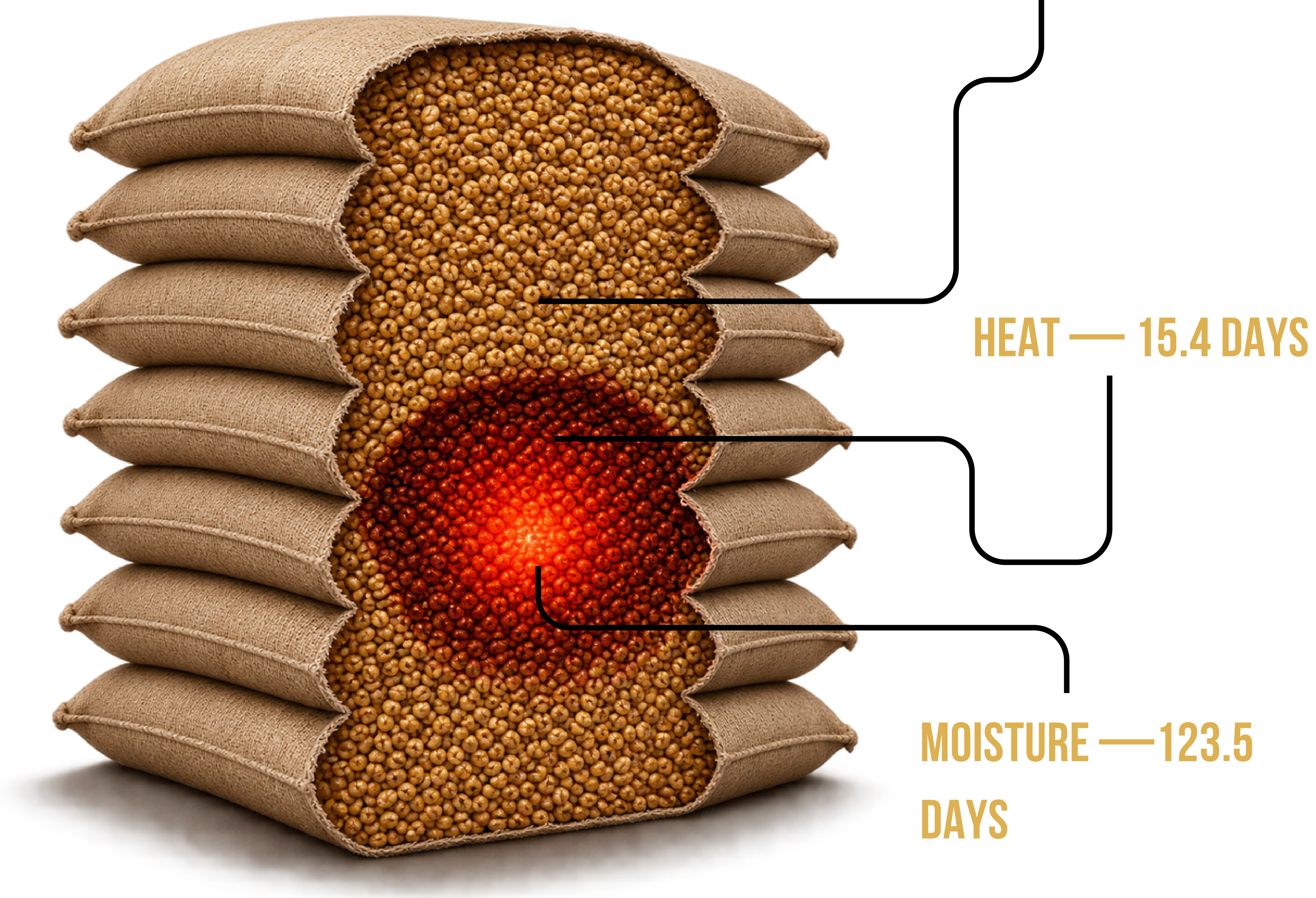
THE SIGNALS TRAVEL AT DIFFERENT SPEEDS

CO_2 escapes through **the air gaps between kernels.** Heat has to conduct through the kernels themselves — and grain is an excellent insulator.

3

SO GAS LEADS BY ~15×

Over the same 0.4	Arrives in
CO_2	**0.9 days**
Heat	15.4 days
Moisture	123.5 days



1

A compost heap gets warm in the middle and gives off gas. A spoiling pocket is the same event — smaller and slower.

2

The same biological event announces itself as gas about two weeks before it announces itself as warmth.

We dont wait for grain to look BAD. We catch it BREATHING.

THE SOLUTION

FROM "SOMETHING SMELLS" TO "L2 AT 2 M. ACT IN 6 DAYS."

1

SENSE

Interstitial CO_2, humidity and temperature at **9 ports**, three depths on three lances

2

INTERPRET

Humidity becomes **water activity**; water activity plus temperature becomes a **growth forecast**

3

DECIDE

Composite risk score, ranked ports, an action card **with a deadline**

4

VERIFY

Post-treatment signature decay **confirms** the intervention actually worked

1

Current practice leaves this loop open at both ends — nothing today verifies a treatment worked, so failures surface only when infestation returns.

2

Output is not an alarm. It is a **port, a reason, and a deadline.**

detects the RESPIRATORY SIGNATURE of spoilage

THE BIOLOGY

IT'S NOT MOISTURE
IT'S WATER ACTIVITY.

WHY MAIZE

Major stored cereal in the PACS and godown system. Published ASAE D245 sorption coefficients exist for shelled corn, so our conversion is a standard — not invented. It is also what is in our jars: **one commodity validated properly, rather than four claimed loosely.**

THE CORE IDEA

Not all water in a kernel is usable — some is chemically bound to starch and protein, and fungi cannot touch it. Water activity is the free fraction, 0 to 1.

MOISTURE (wb)	TEMP	WATER ACTIVITY	MEANING
12%	30°C	**0.657**	SAFE — growth impossible
14%	25°C	0.774	safe
16%	30°C	**0.875**	ACTIVE GROWTH
18%	25°C	0.926	rapid
20%	30°C	**0.963**	SEVERE

TIME TO VISIBLE MOULD AT 30 °C

aw	ONSET
0.95	6.7 days
0.9	10.8 days
0.87	17.4 days
0.85	28.9 days
0.83	86.8 days

0.82 — GROWTH FLOOR

Below this A. flavus physically cannot pull water out of the grain. Not "slow" — cannot.

0.85 — THE TOXIN LINE

Growth starts at 0.82, but aflatoxin production needs conditions wetter still.

25–35 °C — THE TOXIN BAND

Worst at 25–30. Exactly an Indian godown. (Growth range 12–43 °C, fastest at 33.)

1 A pocket only has to cross about 15 % moisture. Then the clock starts — and the clock is what we report.

2 Two batches at the same moisture behave differently at different temperatures, because temperature changes how much water is free.

MOISTURE IS THE TOTAL. AW IS THE SPENDABLE PART.

THE BIOLOGY

YOU CAN STOP THE FUNGUS.
YOU CANNOT UNMAKE THE TOXIN.

Aflatoxin is a liver carcinogen. Drying does not destroy it. Cooking does not destroy it — it is heat stable.

Once aflatoxin is in a consignment, that grain's fate is decided.

THE ONLY USEFUL WINDOW IS BEFORE

The only intervention that changes the outcome happens before the fungus grows.

DIAGNOSIS IS TOO LATE TO ACT ON

A system that tells you grain is already contaminated has told you nothing you can act on.

SO THE PROBLEM IS PREDICTIVE

We forecast the conditions that create toxin — we do not wait to confirm it exists.

1 WE DO NOT MEASURE TOXIN. We predict the conditions that create it, and we say so in every output the system produces.

2 Confirming toxin needs a laboratory assay. A system claiming to read toxin off a humidity sensor would be indefensible.

STATING THE LIMIT IS WHAT MAKES THE REST BELIEVABLE.

CURRENT PRACTICE

THE PROBLEM ISN'T DILIGENCE. IT'S REACH.

HOW INSPECTION WORKS TODAY

An inspector walks the stack looking for staining, mould and insect holes, then pushes a slotted hollow spear into some bags and examines the sample. Careful, skilled work — physically limited in three ways.

1. **IT ONLY REACHES THE OUTSIDE**
   A spear reaches roughly 0.4 m. A godown stack is metres deep. The middle is simply not reachable.

2. **PROBLEMS ARE NOT SPREAD EVENLY**
   Insects and moisture gather in pockets. A clean sample from one spot says nothing about a pocket two metres away.

3. **THE EARLY STAGES ARE INVISIBLE**
   Eggs and larvae develop inside the kernels. A sample with no adult insects can still be heavily infested.

WHY THE CHEMISTRY IS FAILING

RESISTANCE IS DOCUMENTED IN INDIA

Tribolium castaneum at 2.11 to 70.94-fold phosphine resistance (Sci. Reports 13:16497, 2023). Blanket dosing selects for exactly this.

UNTARGETED

You gas the whole stack because you don't know which part is affected

REACTIVE

fumigation kills insects; it does not un-rot grain, and it cannot remove aflatoxin

1 "A spear reaches forty centimetres. Our hotspot is at one point six metres. No amount of diligence gets a spear to that location."

2 This is not a criticism of inspection. It is precisely the reason an interior probe needs to exist.

A PHYSICAL LIMIT, NOT A HUMAN ONE.

THE EVIDENCE

SAME EVENT.
TWO TIMELINES.

HOW THE BASELINE IS MODELLED — AND WHY IT IS FAIR

ASSUMPTION	OUR VALUE	WHY IT IS GENEROUS TO THE BASELINE
Inspection interval	every 14 days	Fortnightly is the generous end — many facilities are monthly or slower
Spear reach	0.4 m	The physical limit of a bag spear into a stacked wall of bags
Finding a pocket in reach	60%	Problems cluster; six in ten is reasonable, not harsh
Surface mould	always found	We assume inspectors never miss visible mould — not once
Musty smell / warm bags	also found	A second detection route we ADDED to help the baseline

1 Our first model had conventional inspection NEVER finding this pocket — arguably what the guidance implies.

2 We thought that looked rigged, so we added smell-and-warmth detection. That is where day 42 comes from. Without it our gap would look much bigger.

WE WEAKENED OUR OWN HEADLINE NUMBER ON PURPOSE.

ARCHITECTURE

FOUR LAYERS.
ONE ANSWER: WHERE, HOW BAD, HOW LONG.

SENSE

9 ports · 3 lances × 3 depths

One shared NDIR analyser across all nine sampling points. SHT31 humidity at ±2 % — because relative humidity at equilibrium IS water activity.

INTERPRET

Isotherm → growth kinetics

Humidity becomes water activity; water activity and temperature feed a cardinal-value growth model giving days-to-onset per zone.

NETWORK

One node per stack · offline-first

Zero external dependencies, zero CDN references. Runs entirely with WiFi off, because rural godown connectivity cannot be assumed.

SEE

Ranked ports · risk map

Every stack colour-coded, every hotspot with a coordinate, a water activity, a deadline and an accumulated dry-matter loss.

1 THE GOTCHA WE DESIGNED AROUND: every NDIR sensor auto-rezeros to the lowest value it sees. Inside grain it never meets fresh air — so ABC must be disabled in firmware, or the sensor erases the signal.

2 Under the World's Largest Grain Storage Plan, 500+ PACS godowns are under construction. None are instrumented today.

CAPACITY IS OUTPACING THE ABILITY TO MONITOR IT.

ARCHITECTURE · DETECTION

WE CHOSE THE EXPLAINABLE DETECTOR.

× NOT A FIXED THRESHOLD

Godown temperature swings seasonally. Any absolute limit either cries wolf in summer or sleeps through winter.

× NOT ALL NINE PORTS TOGETHER

Depth stratification is real and about 1,000 ppm — deep grain is furthest from a venting face. Comparing all nine buried the signal completely.

✓ DEPTH-MATCHED DIFFERENTIAL COMPARISON

Each port is compared only against the other lances at its own depth, on the same analyser. This cancels sensor drift and slow ventilation changes at the same time — and it is the reason a real pocket stands out at all.

THE MEASURED SEPARATION

CLEAN GRAIN PEAKS AT	1.5 Σ
WE ALERT AT	3.0 Σ
A REAL HOTSPOT REACHES	4.2 Σ

Sustained over 3 consecutive readings — one noisy sample is never an alarm.

1 The detector runs on deliberately imperfect data: CO_2 ±25 ppm + 1.5 %, RH ±1.0 %, temperature ±0.15 °C — real datasheet noise, applied to every reading.

2 An explainable detector we can defend beats a black box we cannot. That is an engineering choice, not a shortfall.

EVERY ALERT COMES WITH ITS REASON.

WHAT WE BUILT

NOT A MOCKUP.
YOU CAN BREAK IT YOURSELF.

THE WORKING DIGITAL TWIN

- Explicit finite-difference solver — CO_2, heat and moisture through the grain bed, 4.0 × 2.5 m stack at 10 cm cells
- Modified Chung–Pfost isotherm (ASAE D245) and a cardinal-value predictive mycology model
- Virtual sensors carrying datasheet-accurate noise, so the detector is proven on imperfect data
- A conventional-inspection arm running in parallel on the same injected event
- Self-checking — headless assertions must pass before every session

PHYSICAL VALIDATION — STATED PRECISELY

CLAIM	STATUS
Two sealed maize jars — one at storage moisture, one deliberately wetted	YES
SHT3x reading headspace on the same ESP32 firmware built for the field probe	YES
Reading fed into the SAME decision-engine code as the simulation	YES
Dry jar below the 0.82 growth floor; wet jar above it, with a predicted onset	YES
“We watched grain spoil” — sealed 15 hours, fungus needs days	NO
“We measured CO_2” — sensor not sourced yet	NO

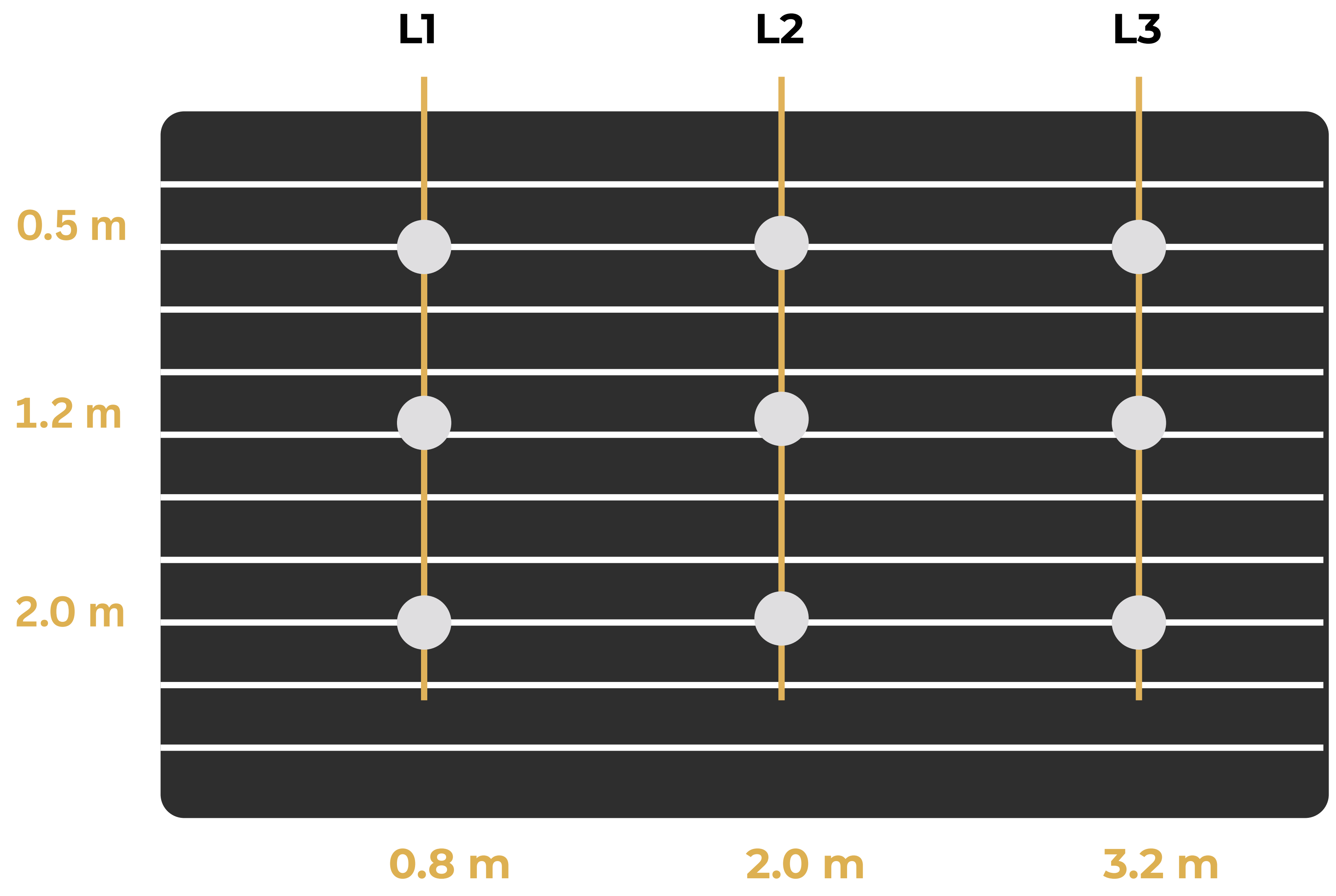
1. The jars demonstrate a measured water-activity difference through real hardware and real code. They do not demonstrate observed spoilage or a CO_2 signal.

2. Being precise about that boundary is stronger than blurring it — and it is the difference between a team that gets believed and one that gets caught.

STATE THE BOUNDARY BEFORE THEY FIND IT.

DEPLOYMENT

THREE LANCES.
NINE EYES.



4.0 m stack face · 3 lances × 3 depths = 9 sampling ports

SHARED ANALYSER

Ports feed a manifold into a single NDIR instrument — the expensive part serves nine sampling points. A hardware decision, not a software one.

DEPTH IS ACCOUNTED FOR

1.2 m is the point furthest from any face and naturally reads highest CO_2. Depth-matched comparison cancels exactly that.

NON-DESTRUCTIVE

Perforated probe lance, driven between bag courses. Removable and reusable across stacks.

HARDWARE, COSTED

SCD30 CO_2 ₹5,000 · 3× SHT31 ₹3,000 ·ESP32 ×2 ₹1,200 · TCA9548A ₹250· supply, cable, connectors ₹1,000 → ₹9,000–11,000, inside the ₹19,800 proposal budget

1 All-3.3 V I²C by design — ESP32 GPIO is not 5 V tolerant, and a TCA9548A multiplexer solves the SHT31 address collision that would otherwise stop three sensors sharing one bus.

2 COVERAGE RESULT — to be inserted: “Three lances at this spacing detect N % of hotspots within X days.”

ONE ANALYSER. NINE SAMPLING POINTS.

INTERVENTION

DETECT EARLY ENOUGH
AND YOU NEVER NEED THE CHEMICAL.

TIER 1 · AUTOMATIC

AERATE

Drive existing godown aeration over the flagged zone. Reversible, non-toxic, cheap. The system watches CO_2 and water activity fall and confirms it worked. Early detection is what makes this sufficient.

TIER 2 · HUMAN HOURS

TURN / ISOLATE

Restack: pull the affected bags, dry and separate them, return the sound grain. Only possible because you know which bags — without localisation this option does not exist.

TIER 3 · LICENSED OPERATOR

TARGETED FUMIGATION

Insect activity confirmed and the pocket established: fumigate that stack, not the shed. Less chemical, less exposure, less selection pressure for resistance.

THEN · AND NOBODY ELSE DOES

VERIFY

Post-intervention signature decay. If the CO_2 anomaly does not fall, the system escalates rather than assuming success. Current practice never checks at all — which is why failures surface only when infestation returns.

1 Late detection leaves only Tier 3, applied to everything, on a chemical that resistance is already eroding.

2 Early detection resolves most events at Tier 1 — the fumigant becomes the exception rather than the routine.

THE BEST TREATMENT IS THE ONE YOU NEVER NEED.

THANK YOU

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